

DevOps Lab

Documentation

Program 6:

Deploy an Automated Build Pipeline using Docker Hub

The program aims to deploy an automated build pipeline using Docker Hub for continuous image building and distribution. This ensures that every code push to the repository triggers Docker Hub to build and store updated container images.

Project Structure:

```
program_6/  
—>Dockerfile  
—>app.py  
—> requirements.txt
```

Procedure:

Create a folder (ex: program_6)

Step 1: Create a simple **Dockerfile** without any extensions

nano Dockerfile

Create a simple [app.py](#) file

nano [app.py](#)

Create a simple requirements.txt file

nano requirements.txt

Build an image

docker build -t program_6 .

Run the container

docker run -p 5000:5000 program_6

```
1RV24MC062_lubna_tabassum@Inspiron:~$ cd DevOps
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps$ mkdir program_6
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps$ cd program_6
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ nano Dockerfile
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ nano app.py
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ nano requirements.txt
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ docker build program_6 .
ERROR: docker: 'docker buildx build' requires 1 argument

Usage:  docker buildx build [OPTIONS] PATH | URL | -

Run 'docker buildx build --help' for more information
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ docker build -t program_6 .
[+] Building 3.3s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 479B                                0.0s
=> [internal] load metadata for docker.io/library/python:latest    3.1s
=> [internal] load .dockerignore                                    0.0s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/python:latest@sha256:6942ebef735aad5f708 0.0s
=> [internal] load build context                                    0.0s
=> => transferring context: 743B                                     0.0s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> CACHED [3/5] COPY requirements.txt                               0.0s
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt 0.0s
=> CACHED [5/5] CMD ["python app.py"]                             0.0s
```

Step 2: Tag the created image for Docker Hub

`docker tag program_6 <dockerhub-username>/program_6:latest`

```
=> => naming to docker.io/library/program_6                        0.0s
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ docker run -p 5000:5000 program_6
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.3:5000
Press CTRL+C to quit
172.17.0.1 - - [19/Nov/2025 10:17:40] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [19/Nov/2025 10:17:40] "GET /favicon.ico HTTP/1.1" 404 -
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ docker tag program_6 <un>/program_6:latest
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ docker tag program_6 lubna01/program_6:latest^C
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ docker tag program_6 lubna01/program_6:latest
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ docker login -u lubna01

Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit https://app.docker.com/settings

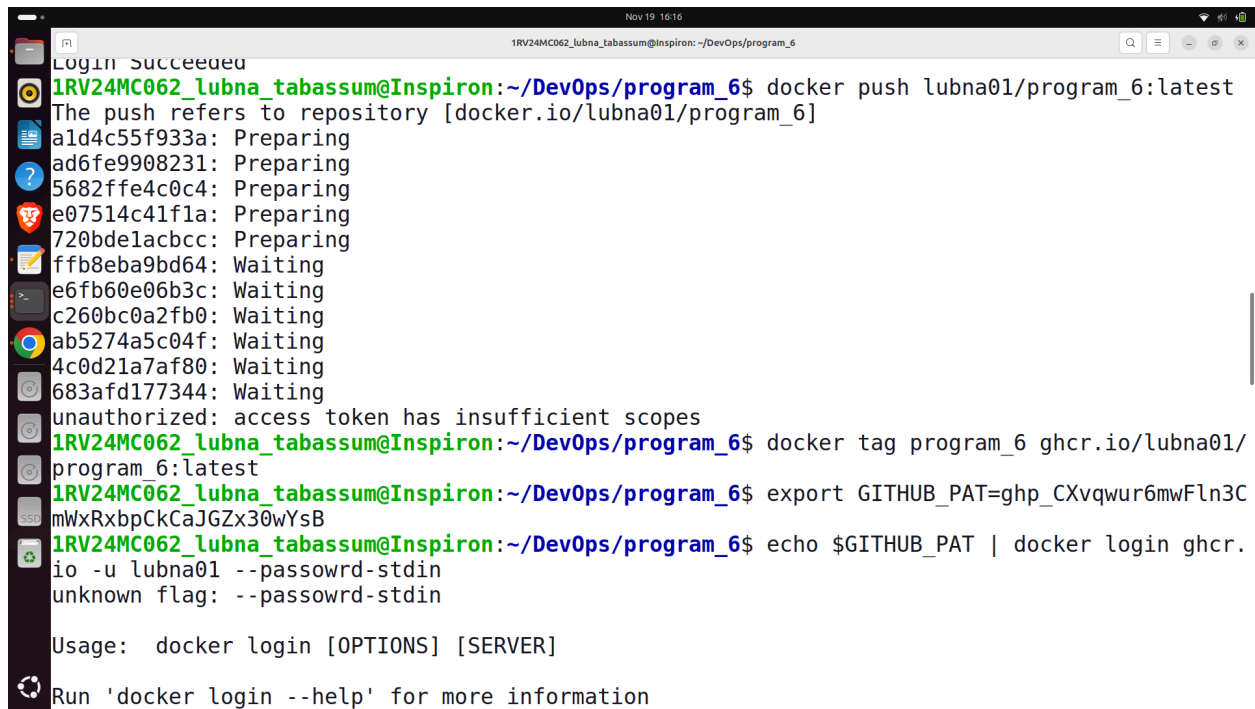
Password:
```

Step 3: Tag the Image for GitHub Container Registry (GHCR)

`docker tag program_6 ghcr.io/github-username/program_6:latest`

Push the Image to GHCR

`echo $GITHUB_PAT | docker login ghcr.io -u github-username --password-stdin`



```
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1RV24MC062_lubna_tabassum@Inspiron: ~/DevOps/program_6
Login Succeeded
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ docker push lubna01/program_6:latest
The push refers to repository [docker.io/lubna01/program_6]
a1d4c55f933a: Preparing
ad6fe9908231: Preparing
5682ffe4c0c4: Preparing
e07514c41f1a: Preparing
720bde1acbcc: Preparing
ffb8eba9bd64: Waiting
e6fb60e06b3c: Waiting
c260bc0a2fb0: Waiting
ab5274a5c04f: Waiting
4c0d21a7af80: Waiting
683afd177344: Waiting
unauthorized: access token has insufficient scopes
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ docker tag program_6 ghcr.io/lubna01/
program_6:latest
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ export GITHUB_PAT=ghp_CXvqwur6mwFln3C
mWxRxbpCkCaJGZx30wYsB
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ echo $GITHUB_PAT | docker login ghcr.
io -u lubna01 --password-stdin
unknown flag: --password-stdin

Usage:  docker login [OPTIONS] [SERVER]

Run 'docker login --help' for more information
```

Step 4: Push source code to GitHub

```
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ git push -u origin main
fatal: 'origin' does not appear to be a git repository
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ git remote -v
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ git push -u origin main
fatal: 'origin' does not appear to be a git repository
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ git remote add origin https://github.
com/lubnatabassum-ux/program_6.git
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ git push -u origin main
Username for 'https://github.com': lubnatabassum-ux
Password for 'https://lubnatabassum-ux@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 733 bytes | 733.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
remote: This repository moved. Please use the new location:
remote: https://github.com/lubnatabassum-ux/program_6.git
```

```
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ git remote -v
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ git push -u origin main
fatal: 'origin' does not appear to be a git repository
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ git remote add origin https://github.
com/lubnatabassum-ux/program_6.git
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$ git push -u origin main
Username for 'https://github.com': lubnatabassum-ux
Password for 'https://lubnatabassum-ux@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 733 bytes | 733.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
remote: This repository moved. Please use the new location:
remote: https://github.com/LubnaTabassum-ux/program_6.git
To https://github.com/lubnatabassum-ux/program_6.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_6$
```

Output

The screenshot shows a web browser window displaying a GitHub repository page for a user named 'LubnaTabassum-ux'. The repository is named 'program_6' and is public. The page layout includes a header with navigation links (Code, Issues, Pull requests, Actions, Projects, Wiki, Security, Insights, Settings) and a search bar. Below the header, the repository name 'program_6' is displayed with a 'Public' badge. A table lists the initial commit for the repository, showing files 'Dockerfile', 'app.py', and 'requirements.txt' committed 24 minutes ago. A section titled 'Add a README' prompts the user to help people understand the project by adding a README file. On the right side, there are sections for 'About' (No description, website, or topics provided), 'Releases' (No releases published), 'Packages' (No packages published), and 'Languages' (Dockerfile 71.4%, Python 28.6%). A 'Suggested workflows' section is also visible, featuring a 'Publish Docker Container' workflow with a 'Configure' button.

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github.com/LubnaTabassum-ux/program_6

LubnaTabassum-ux / program_6

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

program_6 Public

main 1 Branch 0 Tags

Go to file Add file Code

LubnaTabassum-ux Initial commit d990b7c · 24 minutes ago 1 Commit

Dockerfile	Initial commit	24 minutes ago
app.py	Initial commit	24 minutes ago
requirements.txt	Initial commit	24 minutes ago

README

Add a README

Help people interested in this repository understand your project by adding a README.

Add a README

About

No description, website, or topics provided.

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Languages

Dockerfile 71.4% Python 28.6%

Suggested workflows

Based on your tech stack

Publish Docker Container

Build, test and push Docker image to GitHub Packages.

Configure